

August 29, 2024

TECHNOLOGY/SEMICONDUCTORS & COMPONENTS

Stock Rating:

OUTPERFORM

12-18 mo. Price Target \$150.00
NVDA - NASDAQ \$125.61

3-5 Yr. EPS Gr. Rate NA
52-Wk Range \$140.76-\$39.23
Shares Outstanding 24,848.0M
Float 23,854.0M
Market Capitalization \$3,090,006.0M
Avg. Daily Trading Volume 359,857,856
Dividend/Div Yield \$0.04/0.03%
Book Value \$2.34
Fiscal Year Ends Jan
2025E ROE 116.6%
LT Debt \$8,461.0M
Preferred NA
Common Equity \$58,157M
Convertible Available Yes

EPS Diluted	Q1	Q2	Q3	Q4	Year	Mult.
2024A	1.09	2.70	4.02	5.16	12.96	9.7x
2025E	6.12A	0.68A	0.73	0.78	2.80	44.9x
Prior (E)	--	0.62	0.69	0.77	2.70	NM
2026E	0.82	0.87	1.00	1.08	3.77	33.3x
Prior (E)	0.84	0.91	0.98	1.05	--	NM
2027E	--	--	--	--	4.51	27.9x
Prior (E)	--	--	--	--	4.43	NM

EPS are non-GAAP

NVIDIA Corp.

Blackwell as Planned

SUMMARY

NVDA reported upside results/outlook Wed. F2Q (Jul.) sales/EPS 5%/6% ahead. F3Q (Oct.) top-line outlook 3% above Street. Upside led by DC AI for CSP/enterprise. Mgmt. allayed fears of Blackwell design snag, reiterating F4Q (Jan.) production ramp. CoWoS-L supply constraint concerns appear overblown. Hopper expected up 2H/1H led by H200. Mgmt. does not see 4Q "air pocket" some have feared. Mgmt. clarified that NVDA does not plan to be an integrator, instead supplying components to ODMs/OEMs for rack-scale integration. We expect air-cooled NVL36 to constitute most flagship GB200 deployments next year. China DC up Q/Q (>10% by our estimate). GM declined 330bps Q/Q to 75.7% in F2Q, guided 70bps lower in F3Q. FY25 GM outlook near 75%, implies 72% GM in F4Q. We see NVDA best positioned in AI, benefiting from full-stack AI hardware/software. Reiterate Outperform and \$150 target.

KEY POINTS

- DC (87%) up 16% Q/Q (+154% Y/Y) in F2Q led by Hopper and networking. CSP/consumer internet+enterprise contribution 45/55. Inference drove >40% of TTM DC revenue. H200 began ramping F2Q, +40% memory gen/gen. Next-gen models expected to require 20-40x more compute. LLM data requirements grow in proportion to model size. Sovereign AI >\$10B contribution this year (immaterial in CY23). Blackwell Ultra expected late '25, Ruben+Vera due '26, Ruben Ultra '27.
- Networking (14% of DC) up 16% Q/Q (+116% Y/Y) in F2Q. Spectrum-X Ethernet (Switches, DPUs) shipping in volume, expected to reach multi-billion revenue contribution within a year. Ethernet sales doubled Q/Q. Mgmt. claims Spectrum-X performance 1.6x traditional Ethernet scale-out. Mgmt. highlighted xAI using Spectrum-X for 100K node (H100) supercluster. We see InfiniBand share ebbing long-term as open-source Ethernet ecosystem catches up.
- Gaming (10%) up 9% Q/Q (+16% Y/Y) in F2Q driven by console and NB/DT. Channel inventory healthy. Next-gen 50-series Blackwell GPUs launch 2H. Auto (1%) up 5% Q/Q (+37%) as DRIVE platform ramps. Mgmt. secured 80 design wins for Orin and Thor. Auto pipeline \$14B. Pro Vis (2%) up 6% Q/Q (+20% Y/Y) led by AI/graphics. Software \$2B run-rate exiting CY24, up 2x Y/Y.
- FCF \$13.5B (45%) in F2Q. Buybacks/dividends totaled \$7.2B/\$246M. Board approved \$50B share repurchase authorization, adding to existing \$7.5B. Shares split 10-for-1 June 7. Cash/debt \$34.8B/\$8.5B (net cash \$1.06/share). We're raising our CY24/25/26E EPS from \$2.70/\$3.77/\$4.43 to \$2.80/\$3.77/\$4.51.
- Up 154% YTD (SOX +21%), NVDA trades 27x our CY26E EPS vs. five-year average 36x. Leading GM/OM profile, NVDA remains the purest scale play on AI proliferation. Sustained structural growth led by DC AI. NVDA's entrenched DC AI ecosystem is core to GenAI adoption. We remain long-term buyers.

Stock Price Performance



Company Description

Nvidia is a leading supplier of graphics processing units (GPU) for gaming, DC, and auto. GPUs were initially developed for graphics, but are now optimized for accelerating high performance computing and artificial intelligence. NVDA bolstered its data center portfolio with ARM CPU for compute and DPU and InfiniBand for networking.

For analyst certification and important disclosures, see the Disclosure Appendix.

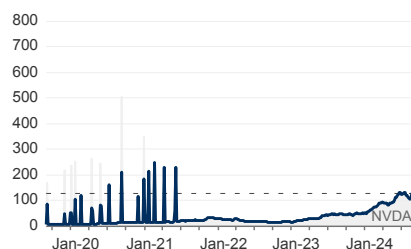
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5-YEAR PRICE PERFORMANCE



Source: Bloomberg

INVESTMENT THESIS

Nvidia has transformed from a graphics company to a premier leading AI computing platform company. GPUs were initially used for graphics in video games and film. Given its parallel processing capabilities and thousands of computing cores, GPU are now optimized for deep learning AI algorithms in datacenters. Through its 2019 Mellanox acquisition, NVDA expanded its DC footprint to high-speed networking DPUs and interconnects. NVDA expanded its DC offerings with Arm-based CPU. We see several structural tailwinds driving sustained outsized top-line growth in high performance gaming, datacenter/AI and autonomous driving vehicles.

BASE CASE ASSUMPTION

- Desktop PC units continue to secularly decline
- Maintains dominant position in Graphics
- GPU AI-accelerator attach maintains steady growth
- Retains dominant share in AI training market
- DC GPU headwinds from US/China export restrictions

UPSIDE SCENARIO

- PC gaming market continues to accelerate off a high base
- Datacenter adoption of GPUs for AI computing continues to accelerate
- Automotive autonomous driving wins ramp faster than expected
- Arm-based CPU displaces x86 in the datacenter
- InfiniBand adoption increases

CATALYSTS

- AI and HPC workloads accelerate datacenter growth
- New consoles and ray-tracing content drive demand for RTX GPUs
- Increased AI accelerator attach in servers
- Large language models such as generative AI

DOWNSIDE SCENARIO

- PCs decline faster than anticipated
- Datacenter growth slows
- Pricing pressure from new AMD and INTC products
- Automotive ADAS faces challenges

PRICE TARGET CALCULATION

Our \$150 price target is based on a 33x P/E multiple of our CY26E EPS of \$4.51. We see several structural tailwinds driving sustained outsized top-line growth including generative AI, DC/AI accelerators and autonomous vehicles. We believe these factors justify our valuation.

KEY RISKS TO PRICE TARGET

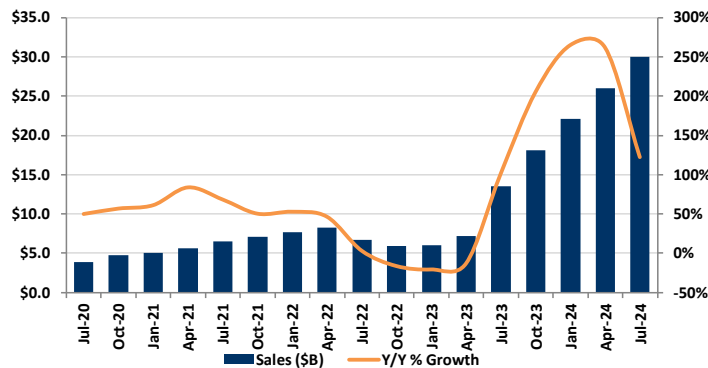
Primary risks to our price target include but are not limited to: 1) slowdown in macroeconomic market or implications from geopolitical factors of tariffs/trade wars; 2) loss of market share from competition with AMD in GPUs; 3) technology manufacturing disruptions at key foundry TSMC; and 4) concentrated DC spending by the top seven hyperscalers.

ESG CONSIDERATIONS *

ESG Rating	C+
ESG Rating Environment	B+
ESG Rating Social	B+
ESG Rating Governance	A

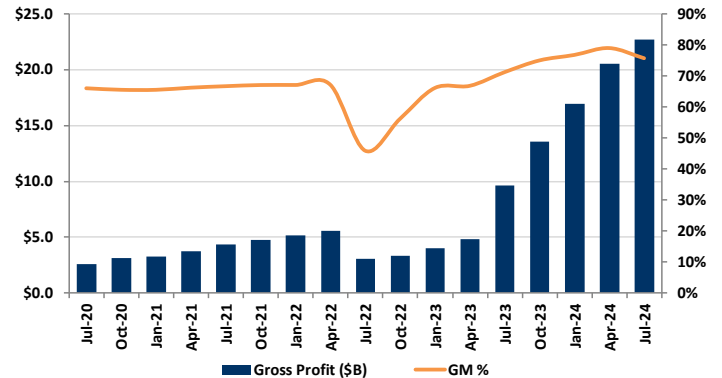
*Environmental/Social/Governance (ESG) scores are courtesy of Refinitiv's ESG product which are designed to transparently and objectively measure a company's relative ESG performance, commitment and effectiveness, based on company-reported data. Refinitiv's ESG ratings are independent of Oppenheimer's stock ratings and are not taken into consideration when assigning a rating. For full details about Refinitiv's ESG scores, please [click here](#)

Exhibit 1: Nvidia Sales; Y/Y Growth %



Source: FactSet and Oppenheimer & Co.

Exhibit 2: Nvidia Gross Profit; GM %



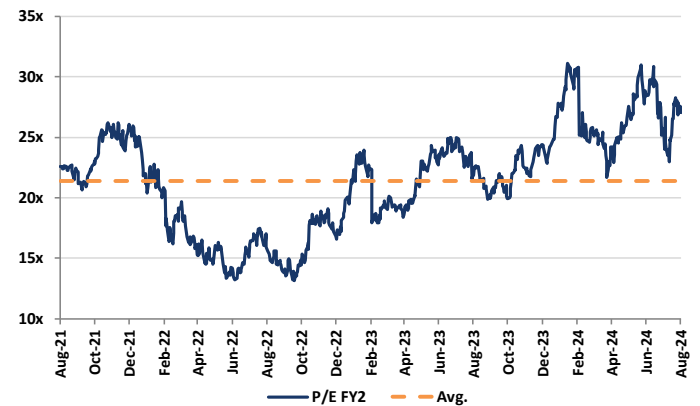
Source: FactSet and Oppenheimer & Co.

Exhibit 3: Nvidia FY2 Forward P/E



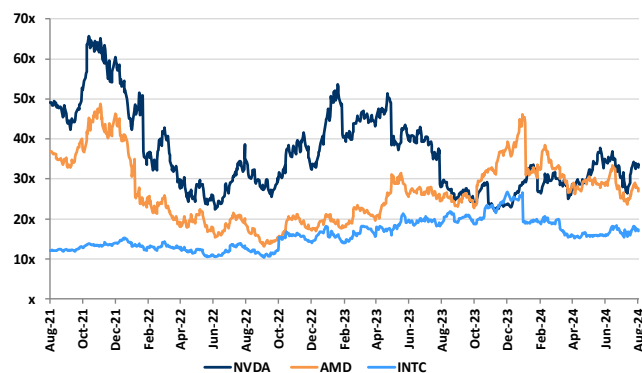
Source: FactSet and Oppenheimer & Co.

Exhibit 4: SOX FY2 Forward P/E



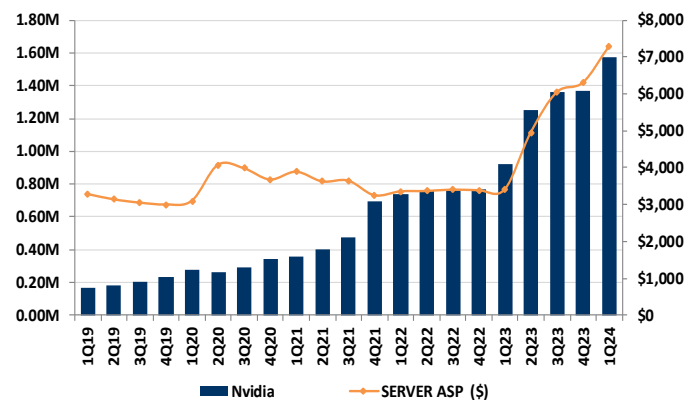
Source: FactSet and Oppenheimer & Co.

Exhibit 5: Nvidia, AMD, INTC FY2 Forward P/E



Source: FactSet and Oppenheimer & Co.

Exhibit 6: Nvidia Server GPU Shipments (M) and ASP (\$)



Note: ASP is for GPU silicon only
Source: IDC and Oppenheimer & Co.

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Rating	Ticker	Name	Price 8/28/2024	Price Target	Mkt Cap	Total Cash	Net Cash	Ent. Value	Net Cash Per Share	EV Per Share	Cash Flow Per Share	Book Per Share	Tang. Book Per Share	Avg Day Vol (M)	Short Int (M)	SI to Float	Days to Cover	Div Yield
Perform	AMD	Advanced Micro Devices	\$157.79	NA	\$258,302	\$5,340	\$3,621	\$254,681	\$2.21	\$155.58	\$1.19	\$34.54	\$7.41	60.6	43.3	3%	0.8	0.0%
Perform	INTC	Intel	\$21.36	NA	\$91,122	\$29,273	-\$23,756	\$114,878	-\$5.57	\$26.92	\$2.72	\$28.22	\$20.76	47.4	91.2	2%	1.5	2.4%
Not Covered	MU	Micron	\$107.89	NA	\$119,650	\$8,379	-\$5,556	\$125,206	-\$5.01	\$112.90	\$4.86	\$39.88	\$38.84	18.5	23.2	2%	1.0	42.6%
Outperform	NVDA	NVIDIA	\$128.50	\$150	\$3,192,968	\$34,800	\$26,339	\$3,166,629	\$1.06	\$127.44	\$1.64	\$2.34	\$2.12	453.0	275.2	1%	0.7	0.0%
Not Covered	STM	ST Micro	\$31.48	NA	\$28,429	\$6,285	\$3,199	\$25,230	\$3.54	\$27.93	\$6.12	\$18.85	\$18.52	3.5	6.3	NA	1.6	1.0%
Perform	TXN	Texas Instruments	\$209.39	NA	\$192,429	\$9,688	-\$4,203	\$196,632	-\$4.57	\$213.96	\$7.09	\$18.73	\$13.98	5.4	20.3	2%	3.8	2.6%

Rating	Ticker	Name	CY EPS		P/E		P/E (Ex-Cash)		FCF Yield	Price/ Cash	Price/ Book	Price/ Tang.Book	CY Sales		P/S		EV/Sales	
			24E	25E	24E	25E	24E	25E					24E	25E	24E	25E	24E	25E
Perform	AMD	Advanced Micro Devices	\$3.41	\$5.33	46.2x	29.6x	45.6x	29.2x	0.8%	NM	4.6x	21.3x	\$25,716	\$32,112	10.0x	9.0x	9.9x	8.9x
Perform	INTC	Intel	\$0.21	\$0.90	NM	23.7x	NM	29.9x	12.7%	NM	0.8x	1.0x	\$52,056	\$53,994	1.8x	3.2x	2.2x	3.6x
Not Covered	MU	Micron	\$4.06	\$10.72	26.6x	10.1x	27.8x	10.5x	4.5%	NM	2.7x	2.8x	\$29,578	\$40,647	4.0x	4.7x	4.2x	4.7x
Outperform	NVDA	NVIDIA	\$2.80	\$3.77	45.9x	34.1x	45.5x	33.8x	1.3%	NM	54.9x	60.7x	\$124,779	\$172,411	25.6x	13.4x	25.4x	13.3x
Not Covered	STM	ST Micro	\$1.68	\$2.29	18.8x	13.7x	16.6x	12.2x	19.5%	8.9x	1.7x	1.7x	\$13,406	\$14,749	2.1x	4.7x	1.9x	4.7x
Perform	TXN	Texas Instruments	\$5.16	\$6.37	40.5x	32.9x	41.4x	33.6x	3.4%	NM	11.2x	15.0x	\$15,748	\$17,972	12.2x	8.7x	12.5x	8.8x
AVERAGE					35.6x	24.0x	35.4x	24.9x	7.0%	8.9x	12.6x	17.1x	\$43,547	\$55,314	9.3x	7.3x	9.3x	7.3x
MEDIAN					40.5x	26.7x	41.4x	29.5x	3.9%	8.9x	3.6x	8.9x	\$27,647	\$36,380	7.0x	6.7x	7.1x	6.7x

Estimates for "Not Covered" Companies come from FactSet.

Source: FactSet, Company Reports and Oppenheimer & Co. Inc.

NVIDIA Oppenheimer & Co.																
\$Mill. Except per Share Data																
Income Statement FY Ends January From Continuing Operations	F2024				F2024	F2025E				F2025E	F2026E				F2026E	F2027E
	April Q1	July Q2	Oct Q3	Jan Q4	Year	April Q1	July Q2	Oct Q3-E	Jan Q4-E	Year	April Q1-E	July Q2-E	Oct Q3-E	Jan Q4-E	Year	Year
Net Sales	\$7,192	\$13,507	\$18,120	\$22,103	\$60,922	\$26,044	\$30,040	\$32,517	\$36,179	\$124,779	\$38,137	\$40,193	\$45,368	\$48,712	\$172,411	\$206,664
QoQ Chg	19%	88%	34%	22%		18%	15%	8%	11%		5%	5%	13%	7%		
YoY Chg	-13%	101%	206%	265%	126%	262%	122%	79%	64%	105%	46%	34%	40%	35%	38%	20%
COGS	2,390	3,893	4,537	5,144	15,964	5,484	7,311	8,141	10,130	31,066	10,678	11,254	12,703	13,639	48,275	59,933
Gross Profit	4,802	9,614	13,583	16,959	44,958	20,560	22,729	24,375	26,049	93,713	27,459	28,939	32,665	35,073	124,136	146,731
Operating Expenses																
R&D	1,339	1,428	1,581	1,747	6,095	1,981	2,241	2,400	2,560	9,182	2,610	2,660	2,710	2,750	10,730	12,020
SG&A	407	405	445	463	1,720	521	551	600	650	2,322	660	670	680	688	2,698	2,957
Operating Expenses	1,750	1,838	2,026	2,210	7,824	2,502	2,792	3,000	3,210	11,504	3,270	3,330	3,390	3,438	13,428	14,977
Pro Forma Op. Income (Loss)	3,052	7,776	11,557	14,749	37,134	18,058	19,937	21,375	22,839	82,209	24,189	25,609	29,275	31,635	110,708	131,754
Net Interest Income	84	120	175	232	611	302	380	350	350	1,382	350	350	350	350	1,400	1,400
Income Before Taxes	3,136	7,896	11,732	14,981	37,745	18,360	20,317	21,725	23,189	83,591	24,539	25,959	29,625	31,985	112,108	133,154
Taxes	423	1,156	1,712	2,142	5,433	3,123	3,365	3,693	3,942	14,123	4,172	4,413	5,036	5,437	19,058	22,636
Net Income	2,713	6,740	10,020	12,839	32,312	15,237	16,952	18,032	19,247	69,468	20,367	21,546	24,589	26,547	93,050	110,518
Avg. Shares Outstanding																
Fully Diluted	2,490	2,499	2,494	2,490	2,493	2,489	24,848	24,808	24,768	24,829	24,728	24,688	24,648	24,608	24,668	24,508
Earnings Per Share																
Non-GAAP EPS	\$1.09	\$2.70	\$4.02	\$5.16	\$12.96	\$6.12	\$0.68	\$0.73	\$0.78	\$2.80	\$0.82	\$0.87	\$1.00	\$1.08	\$3.77	\$4.51
Margin Data																
Gross Margin	66.8%	71.2%	75.0%	76.7%	73.8%	78.9%	75.7%	75.0%	72.0%	75.1%	72.0%	72.0%	72.0%	72.0%	72.0%	71.0%
R&D	18.6%	10.6%	8.7%	7.9%	10.0%	7.6%	7.5%	7.4%	7.1%	7.4%	6.8%	6.6%	6.0%	5.6%	6.2%	5.8%
SG&A	5.7%	3.0%	2.5%	2.1%	2.8%	2.0%	1.8%	1.8%	1.8%	1.9%	1.7%	1.7%	1.5%	1.4%	1.6%	1.4%
Operating Margins	42.4%	57.6%	63.8%	66.7%	61.0%	69.3%	66.4%	65.7%	63.1%	65.9%	63.4%	63.7%	64.5%	64.9%	64.2%	63.8%
Pretax Margin	43.6%	58.5%	64.7%	67.8%	62.0%	70.5%	67.6%	66.8%	64.1%	67.0%	64.3%	64.6%	65.3%	65.7%	65.0%	64.4%
Net Margin	37.7%	49.9%	55.3%	58.1%	53.0%	58.5%	56.4%	55.5%	53.2%	55.7%	53.4%	53.6%	54.2%	54.5%	54.0%	53.5%
Tax Rate	13.5%	14.6%	14.6%	14.3%	14.4%	17.0%	16.6%	17.0%	17.0%	16.9%	17.0%	17.0%	17.0%	17.0%	17.0%	17.0%

Source: Company reports and Oppenheimer & Co. Inc.

NVIDIA Oppenheimer & Co. Inc.								
BALANCE SHEET (\$ MIL)								
FY ENDS JANUARY	Oct-22	Jan-23	Apr-23	Jul-23	Oct-23	Jan-24	Apr-24	Jul-24
Assets								
Cash and cash equivalents	13,143	13,296	15,320	16,023	18,281	25,984	31,438	34,800
Accounts receivable, net	4,908	3,827	4,080	7,066	8,309	9,999	12,365	14,132
Inventories, net	4,454	5,159	4,611	4,319	4,779	5,282	5,864	6,675
Prepaid and other current assets	718	791	872	1,389	1,289	3,080	4,062	4,026
Total Current Assets	23,223	23,073	24,883	28,797	32,658	44,345	53,729	59,633
Property and equipment, net	3,774	3,807	3,740	3,799	3,844	3,914	4,006	4,885
Deposits and other assets	4,507	4,858	5,298	5,736	5,983	5,846	6,100	5,557
Goodwill	4,372	4,372	4,430	4,430	4,430	4,430	4,453	4,622
Deferred income taxes	2,762	3,396	4,568	5,398	5,982	6,081	7,798	9,578
Intangible assets, net	1,850	1,676	1,541	1,395	1,251	1,112	986	952
TOTAL ASSETS	\$40,488	\$41,182	\$44,460	\$49,555	\$54,148	\$65,728	\$77,072	\$85,227
Liability & Shareholders' Equity								
Accounts payable	1,491	1,193	1,141	1,929	2,380	2,699	2,715	3,680
Accrued liabilities	4,115	4,120	4,869	7,156	5,472	6,682	11,258	10,289
Short-term Debt	1,249	1,250	1,250	1,249	1,249	1,250	1,250	0
Total Current Liabilities	6,855	6,563	7,260	10,334	9,101	10,631	15,223	13,969
Long-term operating lease liabilities	798	902	939	1,041	1,091	1,119	1,281	1,304
Long-term debt	9,701	9,703	9,704	8,456	8,457	8,459	8,460	8,461
Other long-term liabilities	1,785	1,913	2,037	2,223	2,234	2,541	2,966	3,336
Total Shareholders' Equity	21,349	22,101	24,520	27,501	33,265	42,978	49,142	58,157
Total Liabilities and Equity	\$40,488	\$41,182	\$44,460	\$49,555	\$54,148	\$65,728	\$77,072	\$85,227
Valuation Ratios								
Profitability Ratios								
Return On Equity	27.3%	39.3%	44.3%	98.0%	120.5%	119.5%	124.0%	116.6%
Return on Avg Equity *	42.3%	34.3%	30.0%	51.0%	79.3%	99.3%	121.7%	128.5%
Return On Assets	14.4%	21.1%	24.4%	54.4%	74.0%	78.1%	79.1%	79.6%
Return On Net Assets	21.3%	31.2%	37.2%	80.4%	111.7%	129.2%	133.6%	134.5%
Return On Sales	24.5%	35.9%	37.7%	49.9%	55.3%	58.1%	58.5%	56.4%
Efficiency Ratios								
Sales/Total Assets	0.6	0.6	0.6	1.1	1.3	1.3	1.4	1.4
A/R Days Sales Out	75.5	57.7	51.8	47.7	41.8	41.3	43.3	42.9
Inventory Turns	2.3	1.6	2.1	3.6	3.8	3.9	3.8	4.4
Days of Inventory	155.8	228.8	175.6	101.0	95.9	93.4	97.3	83.1
Liquidity Ratios								
Current Ratio	3.4	3.5	3.4	2.8	3.6	4.2	3.5	4.3
Quick Ratio	2.7	2.7	2.8	2.4	3.1	3.7	3.1	3.8
Net Working Capital	\$16,368	\$16,510	\$17,623	\$18,463	\$23,557	\$33,714	\$38,506	\$45,664
Long-term Debt/Equity	45.4%	43.9%	39.6%	30.7%	25.4%	19.7%	17.2%	14.5%
Total Debt/Equity	45.4%	43.9%	39.6%	30.7%	25.4%	19.7%	17.2%	14.5%
Book Value and Cash								
Book Value Per Share	\$8.44	\$8.92	\$9.85	\$11.00	\$13.34	\$17.26	\$19.74	\$2.34
Cash & Investments Per Share	\$5.20	\$5.37	\$6.15	\$6.41	\$7.33	\$10.44	\$12.63	\$1.40
Net Cash & Investments Per Share	\$0.87	\$0.95	\$1.75	\$2.53	\$3.44	\$6.54	\$8.73	\$1.06

* Avg Over Last 4 Qtrs

Source: Oppenheimer & Co. and Company Data

Disclosure Appendix

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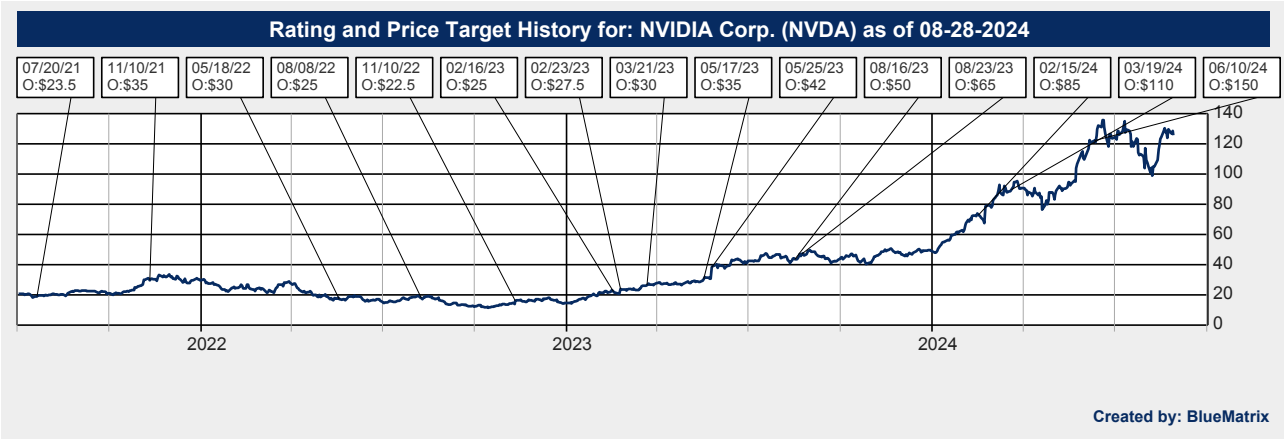
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Advanced Micro Devices (AMD - NYSE, \$146.36, PERFORM)
Intel Corp. (INTC - NASDAQ, \$19.61, PERFORM)
Texas Instruments (TXN - NYSE, \$207.74, PERFORM)

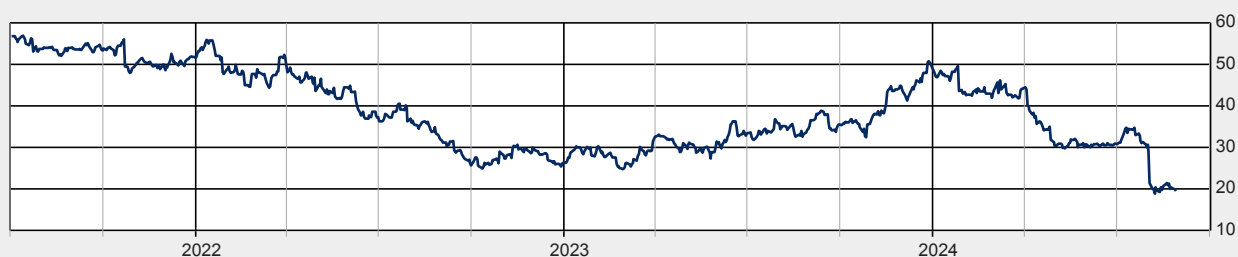


Rating and Price Target History for: Advanced Micro Devices (AMD) as of 08-28-2024



Created by: BlueMatrix

Rating and Price Target History for: Intel Corp. (INTC) as of 08-28-2024



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Rating and Price Target History for: Texas Instruments (TXN) as of 08-28-2024



Created by: BlueMatrix

All price targets displayed in the chart above are for a 12- to 18-month period. Prior to March 30, 2004, Oppenheimer & Co. Inc. used 6-, 12-, 12- to 18-, and 12- to 24-month price targets and ranges. For more information about target price histories, please write to Oppenheimer & Co. Inc., 85 Broad Street, New York, NY 10004, Attention: Equity Research Department, Business Manager.

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Distribution of Ratings/IB Services Firmwide				
Rating	Count	Percent	IB Serv/Past 12 Mos.	
			Count	Percent
OUTPERFORM [O]	431	63.20	185	42.92
PERFORM [P]	251	36.80	99	39.44
UNDERPERFORM [U]	0	0.00	0	0.00

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